ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)
JAIPUR CENTER
Academic Session : 2021-22

SRGBT

for
JEE (MAIN+ADVANCED) 2022
ENTHUSIAST COURSE

Date : 03-10-2021 | **Time : 3.00 Hr** | **CLASS-XII**

Student's Form No. : ..... **Batch :** .....

Student's Name : .....

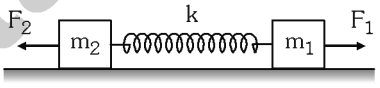
INSTRUCTIONS

1. **PART-1 : Physics** : **(Q. No. 1 to 5)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 6 to 13) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 14 to 18) : **Numerical Type Questions : Marking Scheme (+4, 0)**
2. **PART-2 : Chemistry** : **(Q. No. 19 to 23)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 24 to 31) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 32 to 36) : **Numerical Type Questions : Marking Scheme (+4, 0)**
3. **PART-3 : Mathematics** : **(Q. No. 37 to 41)** : **Single Correct Type Questions : Marking Scheme (+3, –1)**
   (Q. No. 42 to 49) : **One or More Correct Type Questions : Marking Scheme (+4, –2)**
   Partial marks (+1) : (For darkening a bubble corresponding to each correct option, provided no incorrect option darkened)
   (Q. No. 50 to 54) : **Numerical Type Questions : Marking Scheme (+4, 0)**

PART-1 : PHYSICS

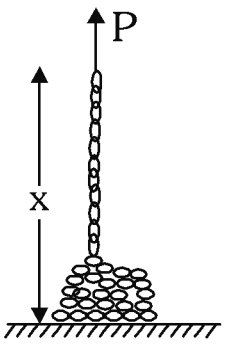
SINGLE CORRECT TYPE (+3, –1)

1. Two blocks of mass m_1 and m_2 are connected by a spring of force constant k. Block of mass m_1 is pulled by a constant force F_1 and other block is pulled by a constant force F_2. Find the maximum elongation in the spring.



(A) $\left[\frac{F_1m_2+m_1F_2}{m_1+m_2}\right]\frac{1}{k}$ (B) $\left[\frac{F_1m_2+m_1F_2}{m_1+m_2}\right]\frac{2}{k}$ (C) $\left[\frac{F_1m_2+m_1F_2}{m_1+m_2}\right]\frac{3}{k}$ (D) $\left[\frac{F_1m_2+m_1F_2}{m_1+m_2}\right]\frac{4}{k}$

2. A chain of length L and mass per unit length ρ is pilled on a horizontal surface. One end of the chain is lifted vertically with a constant velocity v by a variable force P. Which of the following is **incorrect**.



(A) Force P at height L/2 is $\rho\left[g\frac{L}{2}+V^2\right]$

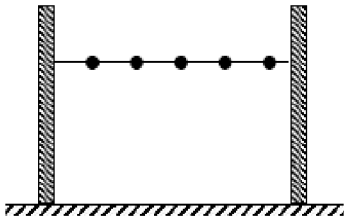
(B) Total work done by force P during lifting the chain is $\frac{\rho gL^2}{2}+\rho V^2L$

(C) Energy lost during lifting the chain is ρLV^2

(D) Energy lost during lifting the chain is $\frac{1}{2}\rho LV^2$

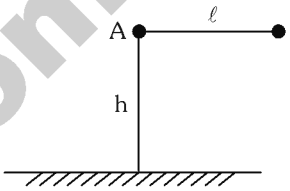
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3. Five identical balls each of mass m and radius r are strung like beads at random and are at rest along a smooth, rigid horizontal thin rod of length L, mounted between immovable supports as shown in the figure. Assume $10r < L$ and that the collision between balls or between balls and supports are elastic. If one ball is struck horizontally so as to acquire a speed v, the average force felt by the support is



(A) $\frac{5mv^2}{L-5r}$ (B) $\frac{mv^2}{L-10r}$ (C) $\frac{5mv^2}{L-10r}$ (D) $\frac{mv^2}{L-5r}$

4. A is a fixed point at a height h above a perfectly inelastic smooth surface. A light inextensible string of length ℓ (ℓ > h) has one end connected to A and other to a heavy particle as shown in figure. The particle is held at the level of A with the string tight and released from rest. The height above the plane, where particle is again instantaneously at rest is S, then which is **correct** ?



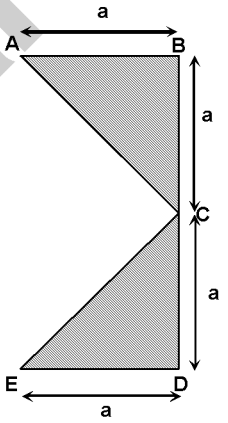
(A) velocity of particle on the surface is $\sqrt{\frac{2gh^2}{\ell}}$

(B) velocity of particle when it leaves surface $\sqrt{\frac{2gh^4}{\ell^3}}$

(C) $S = \frac{h^5}{\ell^4}$

(D) None of these

5. ABCDECA is a planar body of mass m of uniform thickness and same material. The dimensions are as shown in the figure. The moment of inertia of the body about an axis passing through point A and perpendicular to planar body is I_1 and that of about an axis passing through C and perpendicular to planer body is I_2. If I_1/I_2 is k. Find the value of k.



(A) 1 (B) 2 (C) 3 (D) 4

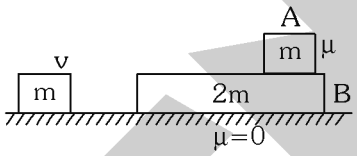
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ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

● Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

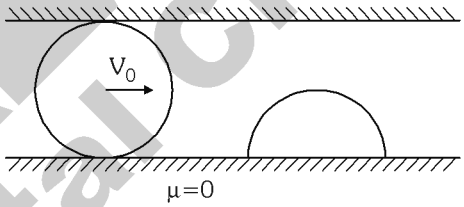
6. A block of mass m is projected with velocity v as shown in figure. The ground is smooth but there is friction between A and B. If collision is elastic then choose correct option : (block B is sufficiently long)



(A) the final common velocity of A and B is $\frac{2v}{3}$ (B) the final common velocity of A and B is $\frac{4v}{9}$

(C) Energy dissipated in friction is $\frac{4}{9}mv^2$ (D) Energy dissipated in friction is $\frac{4}{27}mv^2$

7. A smooth hemisphere of mass m and radius R is at rest. A smooth solid sphere of mass 2m and radius R moving with velocity V_0 between two horizontal smooth surfaces separated by a distance slightly greater than 2R as shown in figure. Solid sphere collides with the hemisphere. If coefficient of restitution is $\frac{1}{2}$, then



(A) The speed of hemisphere after collision is V_0

(B) The speed of solid sphere after collision is $\frac{V_0}{2}$

(C) The loss in kinetic energy of the system is $\frac{1}{4}mV_0^2$

(D) The final kinetic energy of hemisphere is $\frac{1}{4}$th the initial kinetic energy of sphere.

8. A thin hemispherical shell of uniform density and radius R is placed upside down on a rough inclined plane. The shell just starts toppling without slipping. Given that centre of mass of the hemispherical shell lies at a distance of R/2 from its centre, which of the following is true?

(A) Flat surface of the shell should be vertical.

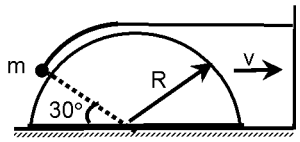
(B) Centre of mass of the shell should be vertically above the lowermost point of the shell.

(C) $\mu > 2$

(D) $\mu > \frac{1}{2}$

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9. A point object of mass m is slipping down on a smooth hemispherical body of mass M and radius R. The point object is tied to a wall with an ideal string as shown. At a certain instant, speed of the hemisphere is v and its acceleration is a. Then speed v_p and acceleration a_P of a particle has value (Assume all the surfaces in contact are frictionless)



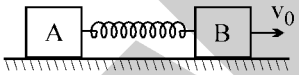
(A) $v_P = v \sin 60°$

(B) $v_P = v$

(C) $a_P = a$

(D) $a_P = \sqrt{\left[\left(\frac{v^2}{R}\right)+\left(a\frac{\sqrt{3}}{2}\right)\right]^2+\left(\frac{a}{2}\right)^2}$

10. Two blocks A and B of the same mass are connected to a light spring and placed on a smooth horizontal surface. B is given velocity v_0 (as shown in the figure) when the spring is in natural length. In the subsequent motion.



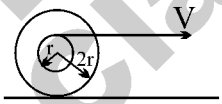
(A) the maximum velocity of B will be v_0

(B) as seen from ground, A can move towards right only

(C) the spring will have maximum extension when A and B both stop

(D) the spring will be at natural length again when B is at rest

11. A bobbin of mass m and moment of inertia I relative to its own axis is being pulled towards right along a horizontal surface by the light string tightly wrapped as shown in figure. There is no slipping on the surface throughout the motion.



(A) If string is pulled by horizontal velocity V, then the angular velocity of bobbin is $\omega = \frac{V}{3r}$

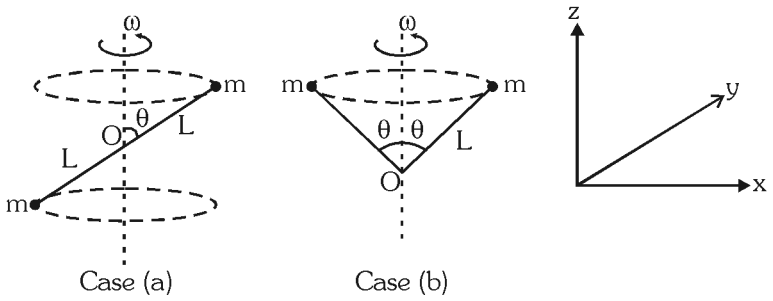
(B) If string is pulled by horizontal velocity V, then the angular velocity of bobbin is $\omega = \frac{V}{2r}$

(C) If string is pulled by horizontal acceleration a, then the tension in string $T = \frac{a}{9}\left(4m+\frac{I}{r^2}\right)$

(D) If string is pulled by horizontal acceleration a, then the tension in string $T = \frac{a}{9}\left(2m+\frac{I}{r^2}\right)$

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12. Two particles connected with massless rod are rotating with constant angular velocity ω about axis as shown in figure.



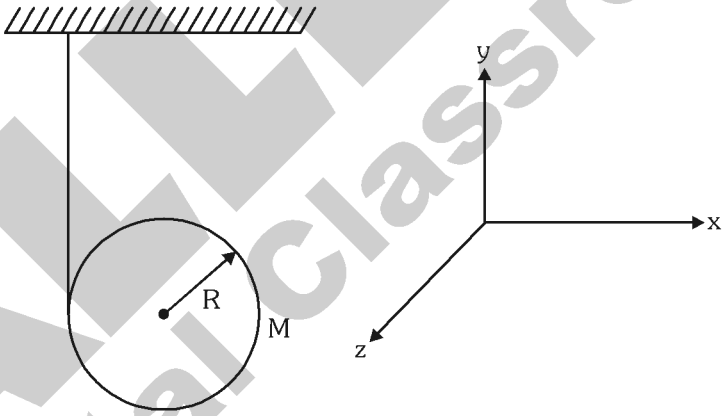
(A) Total angular momentum about O in case (a) and (b) is $2mL^2\sin^2\theta\,\omega\hat{k}$.

(B) Total angular momentum about O is $2mL^2\sin^2\theta\,\omega\hat{k}$ in case (a) only.

(C) Total angular momentum about O is $2mL^2\sin^2\theta\,\omega\hat{k}$ in case (b) only and net torque about O is zero in this case.

(D) In case (a) horizontal torque about O is $m\omega^2L^2\sin 2\theta$

13. A disc of mass M radius R moves under gravity while unwinding string under non slip condition. Mark the correct statements :



(A) Acceleration of the disc is $\frac{2g}{3}$

(B) Rate of $(\vec{L})$ change of angular momentum of disc $\frac{dL}{dt}$ about a point along string is in –k direction.

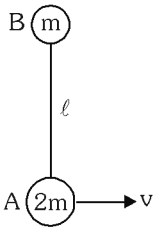
(C) There is no point in the space about which angular momentum of disc is conserved

(D) Rate of change of angular momentum $\frac{dL}{dt}$ of disc about center of mass in $+\hat{k}$ direction.

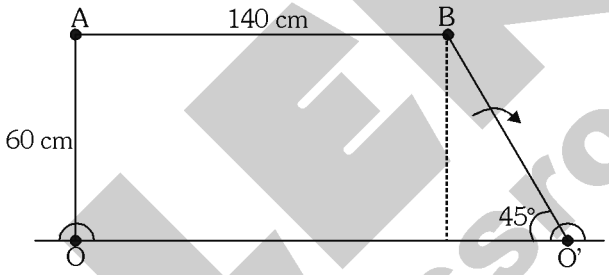
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NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

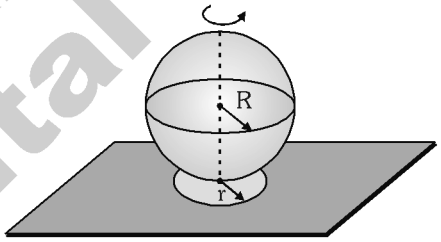
14. Two masses A and B connected with an inextensible string of length ℓ lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in figure. The tension in string during their subsequent motion is $\frac{A}{B}\left(\frac{mv^2}{\ell}\right)$, then find $\frac{A}{B}$?



15. The arm O'B of the linkage has a clockwise angular velocity of 10 rad s^{-1} in the position shown in the figure. What is the velocity of A in ms^{-1} (Assume A & B as rotable joints).



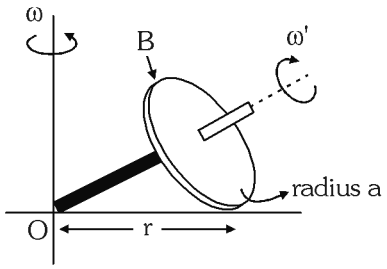
16. A homogeneous sphere of radius R = 10 cm rotating about its vertical diameter is gently released on a hole of radius r_1 = 8.0 cm made in a fixed horizontal slab. Centre of the hole is vertically below the centre of the sphere. The sphere stops in a time interval Δt_1 = 9.0 s. Calculate time interval required to stop the same sphere, if it is placed rotating with the same angular velocity on a hole of radius r_2 = 6.0 cm made in another identical horizontal slab.



17. A smooth fixed ring is kept on a smooth horizontal surface. From a point P of the ring a particle is projected at an angle α to the radius vector at P. If e is the coefficient of restitution between the ring and the particle and the particle will return to the point of projection after two reflections. If $\cot^2\alpha = \frac{A}{e^B} + \frac{C}{e^D} + \frac{E}{e^F}$ then find $\frac{A}{B} + \frac{C}{D} + \frac{E}{F}$?

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18. The shown structure (a rotable disc over perpendicular rod) rolls purely over a horizontal surface. Obtain velocity of position B. (Use : r = 8m, a = 4m & ω = 5 rad/s.)

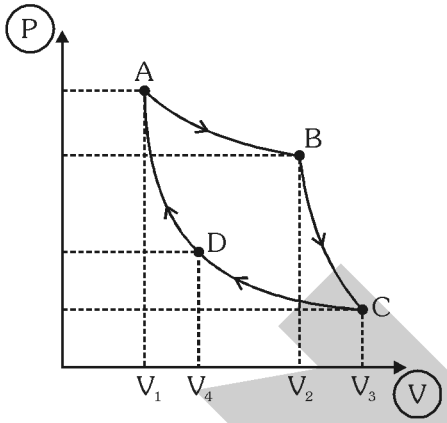


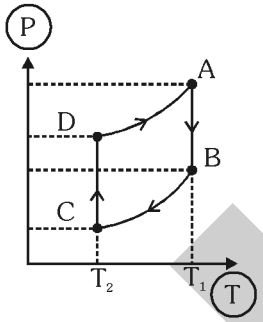
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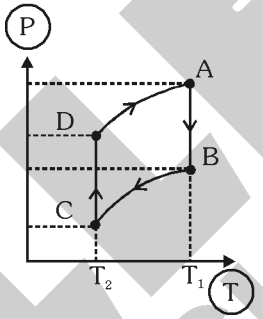
PART-2 : CHEMISTRY

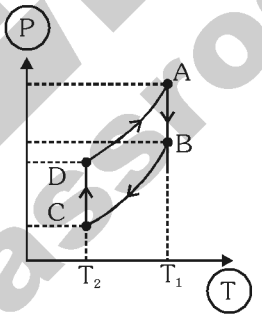
SINGLE CORRECT TYPE (+3, –1)

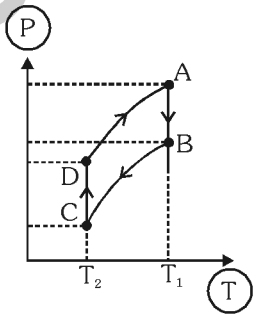
19. For a typical carnot cyclic process as shown in the graph, select the correct P–T graph for the entire cyclic process.



(A)


(B)


(C)


(D)


20. Standard entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 JK^{-1} mol^{-1}, respectively. Standard enthalpy of atomisation of $X_2(g)$ and $Y_2(g)$ are 120 KJ/mole & 180 KJ/mole respectively. Bond enthalpy of X–Y bond (assume XY_3 has three X–Y single bonds) is 120 KJ/mole.

$\frac{1}{2}X_2(g) + \frac{3}{2}Y_2(g) \rightarrow XY_3(g)$, then true about the reaction is :

(A) spontaneous at all temperature below 1250 K
(B) spontaneous at all temperature above 800 K
(C) spontaneous at all temperature below 750 K
(D) spontaneous at all temperature below 1000 K

21. Calculate $|\Delta_f G°|$ for (NH_4Cl, s) at 350 K.
Given : $\Delta_f H°(NH_4Cl, s) = -314.5$ kJ/mol

$S°_{N_2(g)} = 192 JK^{-1}mol^{-1}$; $S°_{H_2(g)} = 130.5 JK^{-1}mol^{-1}$

$S°_{Cl_2(g)} = 223 JK^{-1}mol^{-1}$; $S°_{NH_4Cl(s)} = 94.5 JK^{-1}mol^{-1}$; All given data at 300 K.

$\Delta_r C_P = -20$ J/mol-K ; $ln\left(\frac{350}{300}\right) = 0.15$

(A) 377 kJ/mol
(B) 374 kJ/mol
(C) 183 kJ/mol
(D) 315.5 kJ/mol

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22. Containers A, B, C and D of equal volume V, each having 120g of X_4(linear), X_2, X, X_3(bent) & Tinitial as T, 2T, 3T and 4T respectively. Then find container with maximum $\left\{\frac{\Delta T}{P_{final}}\right\}$ value when all gases compress adiabatically to $\frac{V}{3}$ against constt ext. pressure. (At mass of X = 60 g/mol.) (ignore vibrational degree of freedom).

(A) [A] (B) [B] (C) [C] (D) [D]

23. Which of the following contain correct set of parameters which are both amount and path independent parameters.

(I) Internal energy
(II) Specific enthalpy
(III) Heat exchange at constant pressure condition.
(IV) Molar heat exchange at constant volume condition
(V) Temperature
(VI) Work involved.

(A) Only II, IV & V
(B) All option except VI
(C) All options except III, IV & VI
(D) Only II & V

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

24. 1 mole of a real gas changes its state from – A (2 bar, 3L, 100 K) to state – B (2 bar, 5 L, 200 K) at constant pressure and finally to state – C (3 bar, 10 L, 300 K). If $\Delta U_{BC} = 110$ J and $C_{p,m}$ of gas $= 3\,R = 3 \times 8.3\ JK^{-1}\ mol^{-1}$ then choose the correct option(s) :

(A) $W_{AB} = 830$ J (B) $\Delta H_{AC} = 4600$ J (C) $\Delta U_{AC} = 2200$ J (D) $\Delta U_{AC} = 1770$ J

25. If water vapor is assumed to be a perfect gas, molar enthalpy change for vaporisation of 1 mole of water at 1 bar and 100°C is 41 kJ/mol. Choose the **correct** statement (s)
(Take R = 8.3 J/mol/K)

(A) $\Delta U_{vaporisation}$ of 1 mole of water at 1 bar and 100°C is nearly 37.904 kJ/mol

(B) ($\Delta U \simeq \Delta H$) for conversion of 1 mole of water into ice at 0°C

(C) $\Delta H = 0$, for the conversion of 1 mol of water into ice at 0°C.

(D) ($\Delta H \simeq \Delta U$) for conversion of 1 mole of water into steam 100°C

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26. Which of the following statement(s) is/are incorrect?

(A) The quantities E, H and G have the same dimension

(B) Gibb's free energy of 10 gm ice at 0°C and 1.0 atm is less than the Gibb's free energy of 10 gm water at 0°C and 1 atm.

(C) $\Delta S_{sys} = 0$ for every adiabatic process in a closed system.

(D) For every reversible process, in a closed system, $\Delta S_{sys} = \dfrac{\Delta H_{sys}}{T}$

27. Select the correct statement(s).

(A) In a reversible process ΔG is always zero in a closed system.

(B) In a reversible process ΔS_{univ} is always zero in a closed system.

(C) In a reversible process ΔS_{sys} is always zero in a closed system.

(D) In a reversible process ΔS_{sys} is always zero in an isolated system.

28. Select the correct option(s):

(A) $q = nC_v\, dT$ is applicable to all substances during heating/cooling at constant 'v'.

(B) $q = nC_v\, dT$ is applicable to ideal gas during heating/cooling at constant 'v' only.

(C) $dU = nC_v\, dT$ is applicable for real gas at constant 'v'

(D) $dU = nC_v\, dT$ is applicable for ideal gas at constant 'v' only.

29. Pick out the incorrect statement among the following :

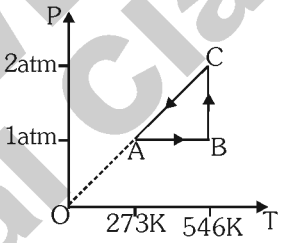
(A) Heat absorb during a change of state (P,V_1,T_1) to (P,V_2,T_2) in a closed system involving ideal gas must be equal to ΔH.

(B) Heat exchanged during a reversible isothermal process between two fixed states is a difinite quantity.

(C) Complete conversion of heat into work is not possible in a reversible isothermal expansion of ideal gas.

(D) Complete conversion of heat into work is not possible in an irreversible isothermal expansion of ideal gas.

30. A cyclic process for 1 mole of monoatomic gas is shown in the P–T diagram. Choose the correct option :



(A) Work done along path $A \to B = -R \times 273$

(B) Change in internal energy along path $BC = \Delta U = \dfrac{3}{2}R \times 273$

(C) For the complete cyclic process $\Delta U = 0$, $\Delta H = 0$, $W = (R \times 273) + \left(2.303\, R \times 273 \log\dfrac{1}{2}\right)$

(D) Heat addition along path $CA = q_{C\to A} = -\dfrac{3}{2}R \times 273$

31. An ideal gas is expanded isothermally from (P_1V_1) to (P_2,V_2) against a constant external pressure. Which of the following is/are correct regarding above process -

(A) If $P_{ext} = P_2$ piston will not stop automatically at the final state.

(B) If $P_{ext} = P_2$ piston will stop automatically at the final state

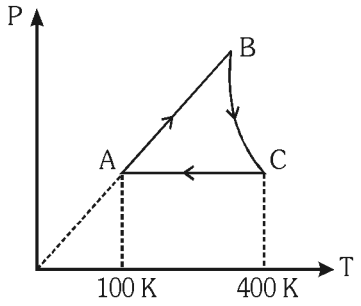
(C) Magnitude of work during expansion can vary in the range of $0 \le |W| \le P_2 (V_2 - V_1)$

(D) Work will be zero when P_{ext} is zero and will be maximum when P_{ext} is equal to pressure of system at final state.

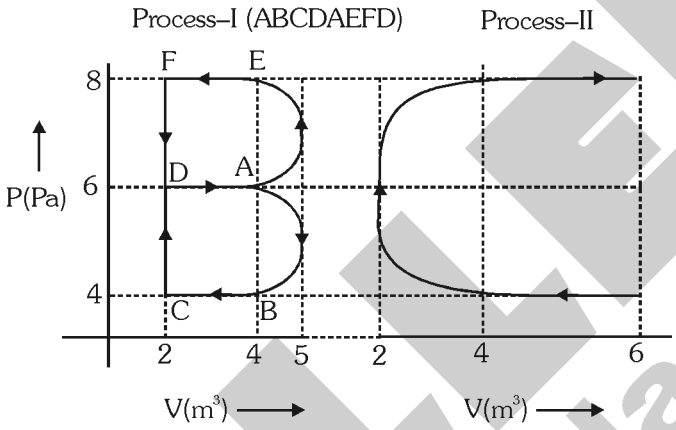
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NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

32. Calculate the magnitude of net work done in the following cycle for 1 mole of an ideal gas in calorie. Where in process BC, PT = const. ($R = 2$ cal $mol^{-1}K^{-1}$)



33. Calculate difference in work done in process–I to process–II.
[Given : $\pi = 3$]

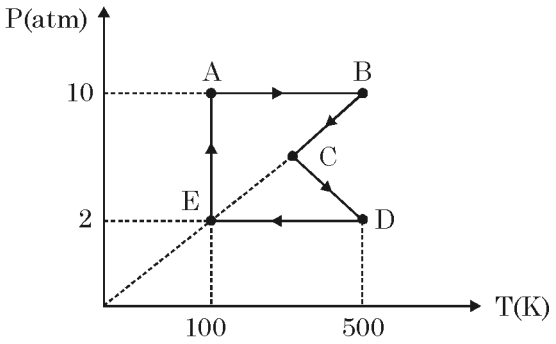


34. For $AB_2(s)$, lattice dissociation enthalpy is 550 kJ mol^{-1} & $\Delta H_{Solution}$ = 50 kJ mol^{-1}. If hydration enthalpies (in kJ/mole) of B^- & A^{2+} ions are in 1 : 4 ratio, then $\frac{|\Delta H_{hydration}|}{10}$ value for cation is ?

35. When 100 mL of 1.0 M HCl was mixed with 100 mL of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of 5.7°C was measured for th beaker and its contents (Expt. 1). Because the enthalpy of neutralization of a strong acid with a strong base is a constant (–57.0 kJ mol^{-1}), this experiment could be used to measure the calorimeter constant. In a second experiment (Expt.2), 100 mL of 1.0 M acetic acid was mixed with 100 mL of 1.0 M NaOH (under identical conditions to Expt.1) where a temperature rise of 5.524 °C was measured. If enthalpy of dissociation of CH_3COOH is 2 KJ/mole, then find the % of CH_3COOH self dissociated in the solution.
(Consider heat capacity of all solutions are 4.2 J g^{-1} K^{-1} and density of all solutions as 1.0 g mL^{-1})

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36.



For 1 mole of ideal gas, consider the cyclic process ABCDEA as shown in the graph through 5 steps (A–B, B–C, ..., E–A).

Select number of correct statements.

(i) At least 3 states have same volume.

(ii) For every step $\frac{V_{final}}{V_{initial}} \neq 1$

(iii) $V_{maximum} = V_B$

(iv) Heat inflow occurs in 3–steps.

(v) Heat outflow occurs in 2–steps.

(vi) Among all steps, maximum $|\Delta U|$ value is observed in A→B and D→E processes.

(vii) $|Q_{BC}| = |Q_{CD}|$

(viii) $\Delta U_{BC} = \Delta U_{CD}$

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PART-3 : MATHEMATICS

SINGLE CORRECT TYPE (+3, –1)

37. If the pair of lines $x^2 + 2xy + \lambda y^2 = 0$ and $\lambda x^2 + 2xy + y^2 = 0$ have exactly one line in common then pair of straight lines representing their other lines is :

(A) $x^2 + 2xy - 3y^2 = 0$ (B) $3x^2 + 2xy - y^2 = 0$

(C) $3x^2 + 8xy - 3y^2 = 0$ (D) $3x^2 + 10xy + 3y^2 = 0$

38. Let S is a circle, from point P outside the circle, tangents are drawn to this circle and points of contact being Q and R. Let A be any point (other than Q and R) on this circle. If a, b, c are length of perpendiculars from A to PQ, PR and QR respectively then

(A) $2c = a + b$ (B) $c^2 = a^2 + b^2$ (C) $2b = a + c$ (D) $c^2 = ab$

39. On the side BC, CA and AB of triangle ABC, semicircles S_1, S_2, S_3 are described externally. If ℓ_1 be the length of common tangent to S_2 and S_3, ℓ_2 be the length of common tangent to S_3 and S_1, ℓ_3 be the length of common tangent to S_1 and S_2. then $(\ell_1\ell_2\ell_3)$ is equal to

(A) $2\Delta s$ (B) Δs (C) $\frac{2\Delta^2}{s}$ (D) $\frac{\Delta^2}{s}$

(where s is semiperimeter and Δ is area of triangle)

40. Circles are drawn to cut both the circles $x^2 + y^2 + 6x + 5 = 0$ and $x^2 + y^2 - 6y + 5 = 0$ orthogonally. If all such circle will pass through two fixed points (x_1, y_1) and (x_2, y_2) then value of $(x_1 + x_2 + y_1 + y_2)$ is equal to

(A) 3 (B) –3 (C) 0 (D) 6

41. Through the point P(2, 3) a pair of perpendicular lines are drawn which meet x-axis point A and B. The locus of incentre of ΔPAB is :

(A) $x^2 + y^2 - 4x - 6y + 13 = 0$ (B) $x^2 + y^2 - 4x - 6y + 9 = 0$

(C) $x^2 - y^2 - 4x - 6y + 9 = 0$ (D) $x^2 - y^2 - 4x - 6y + 13 = 0$

ONE OR MORE THAN ONE CORRECT TYPE (+4, –2)

- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

42. Consider circle $x^2 + y^2 + 6x = 0$ and $x^2 + y^2 - 2x = 0$ which of following are true

(A) Equation of direct common tangents are $x - \sqrt{3}y - 3 = 0$ and $x + \sqrt{3}y - 3 = 0$

(B) Triangle formed by common tangents of two circle is an equilateral triangle

(C) Length of direct common tangent of two circle is $2\sqrt{3}$

(D) Two circles intersect each other at 2 real and distinct points

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43. Tangents are drawn to the circle $x^2 + y^2 = 4$ from the point P(–4, 0). Points of contact of these tangent are T_1 and T_2. A rhombus $PT_1P'T_2$ is completed then correct statements is/are

(A) Circumcentre of the triangle PT_1T_2 is (–2, 0)

(B) Ratio of the area of $\Delta PT_1P'$ to that of $\Delta P'T_1T_2$ is 1 : 2

(C) Point P′ lies on circle $x^2 + y^2 = 4$

(D) Area of quadrilateral $PT_1P'T_2$ is $6\sqrt{3}$

44. Circles are drawn through P(2, 9) touching the co-ordinate axes.

(A) Radius of one of the circles is 17

(B) Radius of one of the circles is 5

(C) common chord of two such circles is $x + y - 11 = 0$

(D) common chord of two such circles is $x + y + 11 = 0$

45. Triangle ABC is formed by the lines $7x - y + 3 = 0$, $x + y - 3 = 0$ and $x - 3y - 31 = 0$. Three new lines are drawn parallel to BC, CA, AB passing through A, B and C intersecting each other at P, Q and R respectively then which of the following is correct :

(A) circumcentre of triangle PQR lies on $3x + y = 3$

(B) circumcentre of triangle PQR lies on $x - y = 9$

(C) circumcentre of ΔPQR, orthocentre and incentre of ΔABC are collinear

(D) Area of ΔPQR is 300

46. The line L is drawn through the point (1, 1) to intersect line $L_1 : x + 3y = 1$ and $L_2 : x + 3y = 4$ at P and Q respectively. Line perpendicular to the L from P intersects L_2 at R and line perpendicular to L from Q intersects L_1 at S. If area of parallelogram PSQR is least then which of the following is/are correct :

(A) The equation of line L can be $2x + y - 3 = 0$

(B) The equation of line L can be $x - 2y + 1 = 0$

(C) The equation of line L can be $x + 2y - 3 = 0$

(D) The least area of PSQR is $\frac{9}{5}$

47. Let ABCD be a rectangle inscribed in a circle whose one diameter is along the line $y = x$. If co-ordinates of A and B are (0, 2) and (4, 3) respectively then which of the following is/are correct :

(A) point C is $\left(\frac{21}{5}, \frac{11}{5}\right)$

(B) point C is (6, 4)

(C) point D is $\left(\frac{1}{5}, \frac{6}{5}\right)$

(D) point D is (4, 6)

48. A circle of radius 3 touches the pair of line $\sqrt{3}y^2 - 4xy + \sqrt{3}x^2 = 0$ in first quadrant at A and B. If C is centre of circle and O is origin then which of the following is correct.

(A) point C is $\left(3+\sqrt{3}, 3+\sqrt{3}\right)$

(B) length of AB is $\frac{3\left(\sqrt{3}+1\right)}{\sqrt{2}}$

(C) circumcentre of ΔABC is $\left(\frac{3+\sqrt{3}}{2}, \frac{3+\sqrt{3}}{2}\right)$

(D) Area of quadrilateral OACB = $9(2+\sqrt{3})$

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49. Let lines OA and OB are represented by $x^2 + y^2 + 4xy = 0$ then possible slopes of AB if ΔOAB is an equilateral triangle is (where O is origin)

(A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) 1 (D) –1

NUMERICAL TYPE
(SINGLE/DOUBLE & THREE DIGITS ANSWER INCLUDING DECIMAL) (+4 , 0)

50. A triangle ABC is such that a circle passing through vertex C, centroid G touches side AB at B. If AB = 6, BC = 4 and AC = ℓ then [ℓ] is [where [.] G.I.F.]

51. Let 3 lines L_1, L_2, L_3 belonging to family $x-2y+6+\lambda(x-y+2)=0$, where λ is a parameter, be internal angle bisectors of ΔABC. If equation of side AB is x + 3y – 4 = 0, then find value of $\left[\frac{4\Delta}{\sum\left(r\cot\frac{A}{2}+a\right)}\right]$ is :

(where Δ is area of ΔABC and [.] is G.I.F.)

52. If circle $x^2+y^2-(2\lambda+1)x+(\lambda-7)y+10=0$ passes through points A and B. If angle between tangents to circle at A and B is 90° then the sum of all possible values of λ is :

53. If $16\ell^2+9m^2=24\ell m+6\ell+8m+1$ and $mx + \ell y = 1$ is tangent to circle $S_1 = 0$ and the equation of of circle $S_2=0$ which touches circle $S_1 = 0$ at (0, 0) and passing through (3, 1) is given by $x^2 + y^2 - ax + by - c = 0$ then value of (3a + b + c) is

54. If $x^2 + y^2 + 2x + 4y + 1 = 0$ and $x^2 + y^2 - 2x - 4y + 1 = 0$ are the equation of two excircles of a triangle then area of triangle is

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SRGBT-11

ENTHUSIAST COURSE

DATE : 03-10-2021

ANSWER KEY

PART-1 : PHYSICS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	C	C	BD	ABC	BC	BD	ABD
Que.	11	12	13	14	15	16	17	18		
Ans.	AC	CD	AB	0.66 to 0.67	6	16	1.83	60		

PART-2 : CHEMISTRY

Que.	19	20	21	22	23	24	25	26	27	28
Ans.	C	C	C	A	A	BC	AB	BCD	BD	ABC
Que.	29	30	31	32	33	34	35	36		
Ans.	ACD	AD	BCD	200	26	33.33	12	2		

PART-3 : MATHEMATICS

Que.	37	38	39	40	41	42	43	44	45	46
Ans.	D	D	D	C	D	ABC	ACD	ABC	ABC	ABD
Que.	47	48	49	50	51	52	53	54		
Ans.	AC	BD	CD	7	4	2	6	2		